

ABHINAY KUMAR NALBAND

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Portfolio: <https://portfolio-rosy-delta-51.vercel.app/>

PROFESSIONAL SUMMARY

Software Engineer with 2+ years of industry experience designing and developing Java-based backend applications, RESTful APIs, and database-driven solutions for enterprise environments. Proficient in building scalable applications using Spring Boot, Hibernate/JPA, and SQL while collaborating in Agile teams throughout the software development lifecycle. Experienced in application testing, debugging, API integration, and delivering maintainable full-stack solutions using version control and collaborative software development practices.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript (ES6+), TypeScript, SQL, C++, C

Backend: Spring Boot, Spring MVC, Spring Security, RESTful APIs, Microservices, Hibernate, JPA, JWT, Maven, Spring Cloud Netflix Eureka

Frontend: Angular, React.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Responsive Web Design

Databases: MySQL, PostgreSQL, Oracle, MariaDB, Relational Database Design

Cloud & DevOps: Docker, Microsoft Azure, Jenkins, GitHub Actions, CI/CD

Testing & API Tools: JUnit, Mockito, Postman, Swagger/OpenAPI, SonarLint, Unit Testing, API Testing

Version Control & Development Tools: Git, GitHub, Bitbucket, Jira

Methodologies: Agile/Scrum, Code Review, Debugging

AI/ML: OpenCV, YOLO Pose, CVAT, scikit-learn, NumPy, pandas, Matplotlib, Jupyter Notebook, RAG, Vector Databases, LLM APIs

WORK EXPERIENCE

Associate Software Engineer

DXC Technologies | Jan 2023 - Aug 2024

- Developed and maintained Java-based backend services using Spring Boot, Spring MVC, Hibernate/JPA, and Maven to implement business logic, RESTful APIs, and database-driven workflows for enterprise applications.
- Designed and integrated RESTful APIs with relational databases by writing SQL queries, implementing repository layers, validating requests, and handling exceptions to support secure and reliable application workflows.
- Built and deployed containerized Spring Boot microservices using Docker and Spring Cloud Netflix Eureka to enable service discovery, improve deployment consistency, and support scalable backend communication.
- Improved application quality by performing unit testing with JUnit and Mockito, validating REST APIs using Postman and Swagger/OpenAPI, and resolving code issues through SonarLint before deployment.
- Collaborated with cross-functional Agile teams using Git, Bitbucket, Jira, and peer code reviews to implement new features, troubleshoot production issues, resolve defects, and deliver production-ready software throughout the SDLC.
- Enhanced existing application modules by implementing new features, debugging production issues, optimizing backend functionality, and supporting continuous improvements based on business and user requirements.

Graduate Research Assistant - Computer Vision & Machine Learning

Texas A&M University | Aug 2025 - May 2026

- Conducted research and experimental analysis on computer vision techniques for aquarium fish and shrimp detection by evaluating image preprocessing methods, background removal algorithms, object detection models, and optical flow approaches to improve detection accuracy and tracking performance.
- Developed and evaluated computer vision workflows using Python, OpenCV, YOLO Pose, CVAT, and GPU-accelerated computing on CUDA-enabled university servers to perform fish detection, body annotation, pose estimation, movement tracking, and dataset preparation for machine learning experiments.
- Collaborated with faculty researchers through weekly research meetings to present experimental findings, document model performance, conduct literature reviews, and contribute to research publications by analyzing results and recommending improvements for subsequent experiments.

Graduate Assistant - Web & Technical Support

Texas A&M University | Dec 2024 - May 2025

- Maintained and enhanced departmental web pages by implementing frontend updates, improving page layouts, resolving content issues, and ensuring accurate, accessible information for students and faculty.
- Collaborated with faculty and university stakeholders to gather requirements, perform website quality assurance, troubleshoot technical issues, and support timely deployment of web content.

Technical Intern

Rakuten PVT. Limited | Jan 2022 - Jun 2022

- Assisted in implementing and enhancing cloud-based application features by updating existing service logic and validating functionality based on business requirements.
- Analyzed and resolved application defects using Jira and Bitbucket while collaborating with senior developers to improve software quality during Agile sprints.
- Performed RESTful API testing using Postman to verify request payloads, response data, and endpoint functionality across development environments.
- Collaborated with development and QA teams to reproduce issues, verify fixes, and support software testing and release activities throughout the SDLC.

ACADEMIC PROJECTS

Micro-Ecommerce Application

Repository: <https://github.com/Abhinay20003/AOS-Ecom>

Tech Stack: Java, Spring Boot, Spring MVC, Spring Security, JWT, Hibernate, JPA, RESTful APIs, Microservices, Spring Cloud Netflix Eureka, Angular, TypeScript, HTML5, CSS3, MySQL, Docker, JUnit, Mockito, Swagger/OpenAPI, Postman, Git

- Developed a full-stack e-commerce application in a team environment, primarily contributing to Angular frontend development and Spring Boot microservices by designing backend services, implementing REST APIs, backend services, and frontend-backend integration workflows.
- Implemented secure user authentication and authorization using Spring Security and JWT-based token management, enabling protected access to customer and administrator workflows with role-based permissions.
- Designed and developed RESTful APIs for product management, cart operations, order processing, checkout workflows, and administrative functions using Spring Boot, Spring MVC, Hibernate/JPA, and MySQL database integration.
- Implemented microservice-based backend architecture with Spring Cloud Netflix Eureka for service discovery and Docker containerization with deployment scripts to ensure consistent application setup and execution.
- Improved application reliability by developing unit tests using JUnit and Mockito and validating API functionality through Swagger/OpenAPI and Postman testing workflows.
- Collaborated with UI/UX designers, developers, and testers using Git-based workflows to integrate features, troubleshoot defects, and deliver a complete full-stack application.

Talk2Campus - AI-Powered Campus Information Assistant

Repository: <https://github.com/PushkarSikharam/Talk2Campus>

Tech Stack: React.js, FastAPI, Python, REST APIs, RAG, ChromaDB, Vector Database, LLM APIs, Web Scraping, WhisperX, Google Maps API, Git

- Developed a full-stack campus information assistant with a React.js frontend and FastAPI backend that enables students, visitors, and new campus users to search university information, access resources, and navigate campus services through an interactive application.
- Designed and implemented a Retrieval-Augmented Generation (RAG) pipeline by collecting and processing university information through web scraping, storing embeddings in ChromaDB vector database, and integrating LLM APIs to provide context-aware chatbot responses.
- Developed and integrated RESTful APIs using FastAPI to connect frontend workflows, retrieval pipelines, chatbot services, and campus information resources while maintaining modular backend architecture.
- Integrated Google Maps API to provide campus navigation functionality and implemented WhisperX-based voice interaction features to improve accessibility for users requiring voice-assisted information access.
- Collaborated with a five-member development team using Git-based workflows while creating and maintaining software documentation including project scope, requirements specification, software design documents, testing documentation, and version updates throughout the development lifecycle.

EDUCATION

Texas A&M University, USA

Master of Science in Computer Science | Aug 2024 – May 2026

GPA: 4.0/4.0

Vignan Institute of Technology and Science, India

Bachelor of Technology in Computer Science

GPA: 8.0/10

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900)
Microsoft | Issued 2023